(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.GAJJARAPU BENNY HIN**

Registration Number: **323129520054**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

IOT-BASED GAS LEAKAGE DETECTION AND EMERGENCY ALERT SYSTEM FOR RESIDENTIAL AND INDUSTRIAL SAFETY

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr.T.Ganesh

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.GAJJARAPU BENNY HIN

Reg.No: 323129520054

Department of MECHANICAL ENGINEERING

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

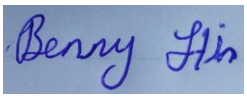
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICAL ENGINEERINGssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

——— << @ >> ———

Student's Declaration

I, **Mr.GAJJARAPU BENNY HIN**, a student of **Bachelor of Technology** Program, Reg. No. **323129520054** of the Department of MECHANICAL ENGINEERING do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.T.Ganesh**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

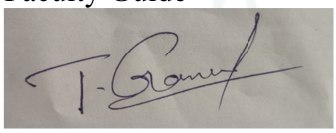
Official Certification

This is to certify that Mr.GAJJARAPU BENNY HIN , Reg. No. **323129520054** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **IOT-BASED GAS LEAKAGE DETECTION AND EMERGENCY ALERT SYSTEM FOR RESIDENTIAL AND INDUSTRIAL SAFETY** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of MECHANICAL ENGINEERING at **Wellfare Institute of Science, Technology and Management**.

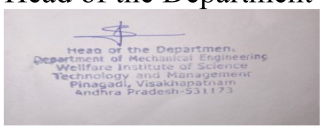
This is accepted for evaluation.

Endorsements

Faculty Guide



Head of the Department



Principal



Certificate from Intern Organization

This is to certify that **Mr.GAJJARAPU BENNY HIN**, Reg. No. **323129520054** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **IOT-BASED GAS LEAKAGE DETECTION AND EMERGENCY ALERT SYSTEM FOR RESIDENTIAL AND INDUSTRIAL SAFETY And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.T.Ganesh**, Assistant Professor of the Department of MECHANICAL ENGINEERING for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an IoT-Based Gas Leakage Detection and Emergency Alert System. Gas leaks in homes, laboratories, and industrial facilities can lead to fires, explosions, health hazards, and significant property damage if not detected promptly. Traditional gas detection systems often rely on local alarms, which may not provide timely notifications when occupants are away. Residential and industrial facilities require intelligent systems capable of continuously monitoring gas concentrations and issuing immediate alerts during emergencies.

The proposed solution is an IoT-Based Gas Leakage Detection and Emergency Alert System that continuously monitors gas levels using IoT sensors and provides real-time emergency notifications. The system offers a centralized platform where gas sensor data is monitored through a secure and user-friendly interface.

By integrating gas sensors such as MQ-series sensors, IoT communication modules, cloud connectivity, and real-time monitoring, the system continuously detects hazardous gas concentrations including LPG, methane, and carbon monoxide. When gas levels exceed predefined safety limits, automated alerts are sent to users and emergency personnel while activating warning alarms and safety mechanisms. Interactive dashboards display live gas concentration, sensor status, alert history, environmental conditions, and system health for continuous safety monitoring.

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1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement an intelligent gas leakage detection simulation using Python that models real-world gas dispersion dynamics.

- To integrate simulated IoT sensors (MQ-5 for LPG, MQ-7 for CO, MQ-4 for Methane) for accurate monitoring with realistic noise and drift.
- To develop an automated alert system that triggers notifications based on predefined safety thresholds (low, medium, high, critical).
- To create a scalable system architecture that supports multiple monitoring zones (residential, industrial, laboratory).
- To implement data visualization and analytics to monitor gas concentration, alert distributions, and daily safety performance.
- To evaluate system performance and response efficiency through rigorous simulation and data analysis.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational IoT-Based Gas Leakage Detection System simulation capable of continuous monitoring and generating instant emergency notifications.
- Facilities can achieve enhanced early leak detection, improved safety, and reduced accident risks.
- Comprehensive analytics dashboards with visualizations of gas concentration trends, sensor activity heatmaps, and alert severity distributions.
- The system architecture supports modular development, scalability for future smart building integration, and efficient resource allocation.
- The system can be extended with advanced features such as automated gas valve shut-off, integration with HVAC systems for ventilation, and direct emergency service dispatch.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted at a leading technology organization focused on Industrial IoT and Smart Safety applications. The organization specializes in bridging the gap between traditional safety infrastructure and intelligent software systems, enhancing facility safety, promoting risk mitigation, and fostering a connected ecosystem. By leveraging emerging technologies such as AI, machine learning, and IoT, the organization aims to augment and upgrade safety infrastructure, enabling property managers and industrial operators to protect their facilities proactively.

2.2 Vision, Mission, and Values

Vision: To combine cutting-edge IoT technology with impactful safety solutions to drive industrial and residential security.

Mission: To support facility managers and homeowners dedicated to improving safety by empowering them with intelligent monitoring tools, thereby creating a widespread network dedicated to hazard prevention and peace of mind.

Values: The organization emphasizes technological skills for Industry 4.0, data-driven decision making, and inclusive access to smart safety technologies for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the mission. Interns are expected to adhere to policies regarding confidentiality, professionalism, active learning, and compliance with ethical guidelines, particularly concerning user safety data and system reliability logs.

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2.4 Organizational Structure

The organization operates under a hierarchical structure including the Board of Directors, Executive Director, Program Managers for Smart Safety Initiatives, Research and Development Team, Administrative Staff, and Interns.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are placed under the guidance of program managers and research teams. Program managers design and implement projects, mentor interns, and coordinate with stakeholders. Research analysts conduct research on IoT safety protocols, prepare reports, and analyze data from sensor deployments.

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CHAPTER 3

PROBLEM ASSESSMENT AND SOLUTION DESIGN

3.1 Problem Analysis

Gas leaks in homes, laboratories, and industrial facilities can lead to fires, explosions, health hazards, and significant property damage if not detected promptly. Traditional gas detection systems often rely on local alarms, which may not provide timely notifications when occupants are away. This leads to significant vulnerabilities, delayed emergency response times, and increased risks for occupants and assets. Users require intelligent systems that can continuously monitor gas concentrations, detect hazardous levels accurately without excessive false alarms, and provide instant notifications for improved safety.

3.2 Key Parameters

Issue to be solved: Inadequate real-time monitoring and delayed response times in traditional gas leakage detection systems.

Target community: Homeowners, apartment residents, laboratory managers, and industrial facility operators.

User needs and preferences: Real-time gas concentration monitoring, instant mobile emergency notifications, comprehensive safety logs, remote monitoring capabilities, and a centralized management dashboard.

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3.3 Requirements Evaluation

3.3.1 Functional Requirements

- The system must simulate MQ-series sensors (MQ-5, MQ-7, MQ-4) to detect LPG, CO, and Methane.
- The system must continuously monitor gas concentrations and apply calibration factors.

- The system must generate automated alerts with varying severity levels (low, medium, high, critical) based on gas concentration thresholds.
- The system must support multiple monitoring zones (e.g., Residential Kitchen, Industrial Warehouse).
- The system must generate visual reports and analytics of gas levels, alert distributions, and daily safety performance.

3.3.2 Non-Functional Requirements

- Reliability: The sensor simulation must include realistic noise and drift to test the robustness of the alert logic.
- Scalability: The architecture must support adding new sensors and zones without significant redesign.
- Performance: The system must process sensor data and generate alerts with extremely low latency.
- Usability: The analytics and reports must be intuitive and actionable for the facility manager.

3.4 Solution Blueprint

The solution blueprint involves a centralized GasDetectionSystem class that manages multiple MonitoringZone objects. Each zone is equipped with GasSensor objects that process gas concentration measurements, and an AlertManager object that manages threshold-based notifications. The system runs a simulation loop representing 15-minute intervals throughout the day, generating realistic gas concentration patterns based on zone type and time of day. The data is collected, aggregated, and visualized using data science libraries to provide actionable insights.

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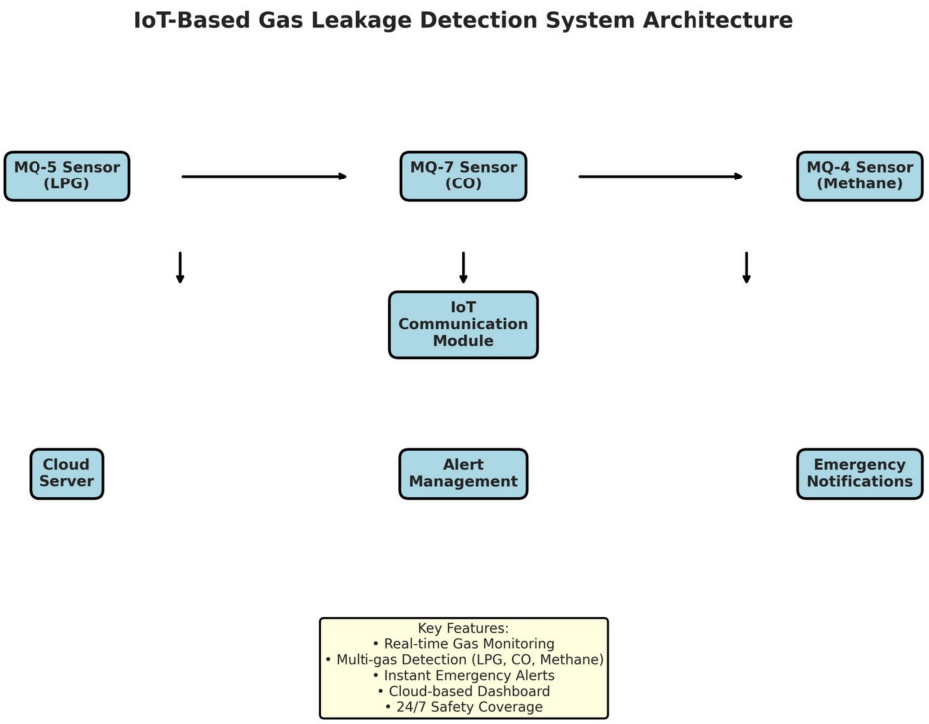


Figure 3.1: IoT-Based Gas Leakage Detection System Architecture

Figure 3.1 illustrates the system architecture. The MQ-series sensors feed data to the IoT Communication module, which transmits it to the Cloud Server. The Alert Management module processes the data against thresholds and triggers Emergency Notifications. Users can access the system through an Analytics Dashboard.

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CHAPTER 4

TECHNOLOGY STACK AND IMPLEMENTATION PLAN

4.1 Technology Stack Selection

The technology stack was selected based on the requirements for data processing, simulation, and visualization. Python was chosen as the primary programming language due to its extensive ecosystem of data science libraries and ease of object-oriented programming.

- Programming Language: Python 3.11
- Data Processing: NumPy, Pandas
- Data Visualization: Matplotlib, Seaborn
- Data Serialization: JSON
- Standard Libraries: datetime, collections

4.2 Project Implementation Plan

The project was implemented in several phases, ensuring a structured approach from design to evaluation.

- Phase 1: Requirement analysis and system design (1 week)
- Phase 2: Development of core sensor classes (GasSensor) (2 weeks)
- Phase 3: Implementation of the main GasDetectionSystem and simulation logic (2 weeks)
- Phase 4: Development of data visualization and reporting modules (1 week)
- Phase 5: Testing, performance evaluation, and bug fixing (1 week)
- Phase 6: Documentation and final report preparation (1 week)

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CHAPTER 5

SOLUTION DEVELOPMENT

5.1 Core Components Implementation

The solution was developed using an object-oriented approach. The system consists of four main classes: GasSensor, AlertManager, MonitoringZone, and GasDetectionSystem.

5.1.1 Sensor Modules

The GasSensor class simulates the behavior of physical MQ-series gas sensors. It includes a `read_gas_level` method that evaluates actual gas concentration and adds a realistic noise factor and sensor drift. The simulation covers MQ-5 for LPG, MQ-7 for Carbon Monoxide, and MQ-4 for Methane.

5.1.2 Alert Management Module

The AlertManager class handles the notification logic. It evaluates the measured gas concentration against predefined safety thresholds (e.g., LPG > 1000 ppm, CO > 35 ppm, Methane > 5000 ppm). It assigns severity levels (low, medium, high, critical) and generates emergency alert messages that would typically be dispatched to the facility manager's mobile device or emergency services.

5.1.3 Gas Detection System Module

The GasDetectionSystem class integrates the zones and orchestrates the simulation. It includes a `simulate_day` method that runs a 24-hour simulation in 15-minute intervals. The simulation generates realistic gas concentration patterns tailored to the zone type (e.g., high baseline with peaks in industrial zones, peaks during cooking in residential kitchens). When a hazardous level is detected, it logs the event and triggers appropriate emergency alerts.

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CHAPTER 6

TESTING AND PERFORMANCE EVALUATION

6.1 Simulation Setup

The system was tested using a 7-day simulation across 5 monitoring zones: ZONE-001 (Residential Kitchen), ZONE-002 (Industrial Warehouse), ZONE-003 (Laboratory), ZONE-004 (Residential Bedroom), and ZONE-005 (Industrial Plant). The simulation generated data for every 15-minute interval, resulting in 96 data points per zone per day.

6.2 Gas Concentration Analysis

The simulation results demonstrated the effectiveness of the real-time monitoring system. By continuously tracking sensor activity, the system successfully identified concentration trends.

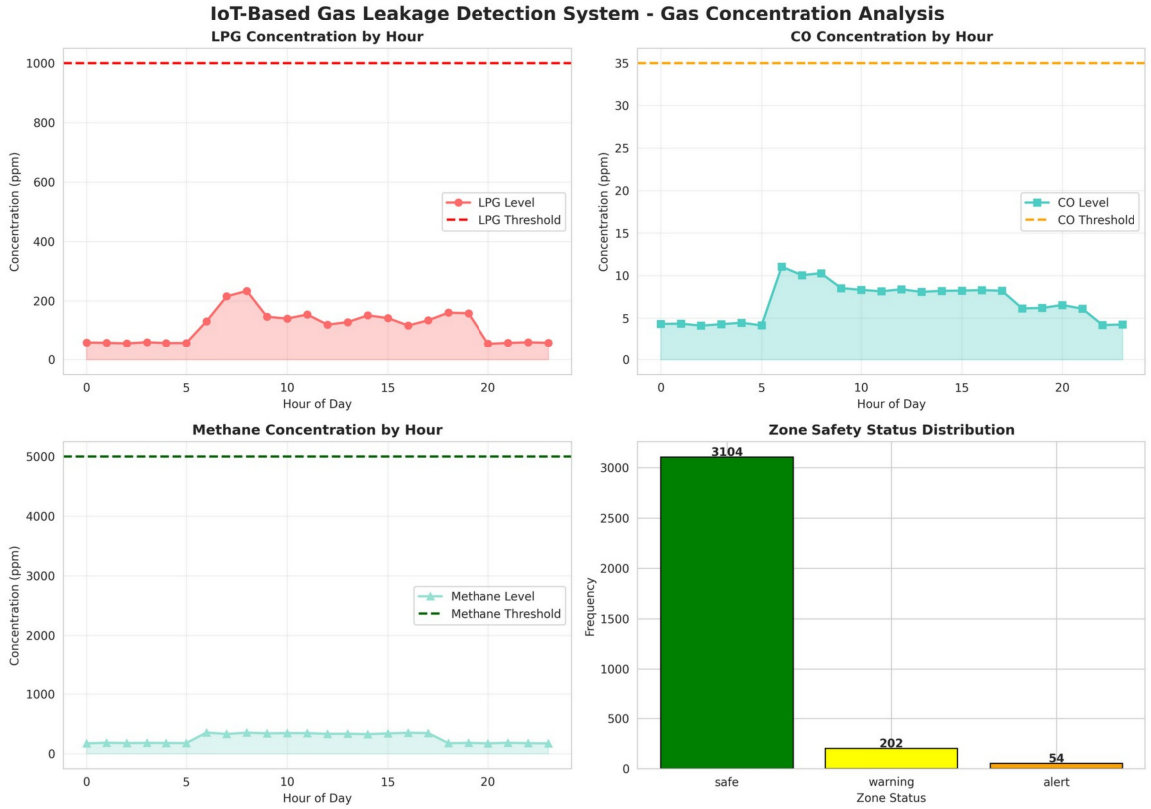
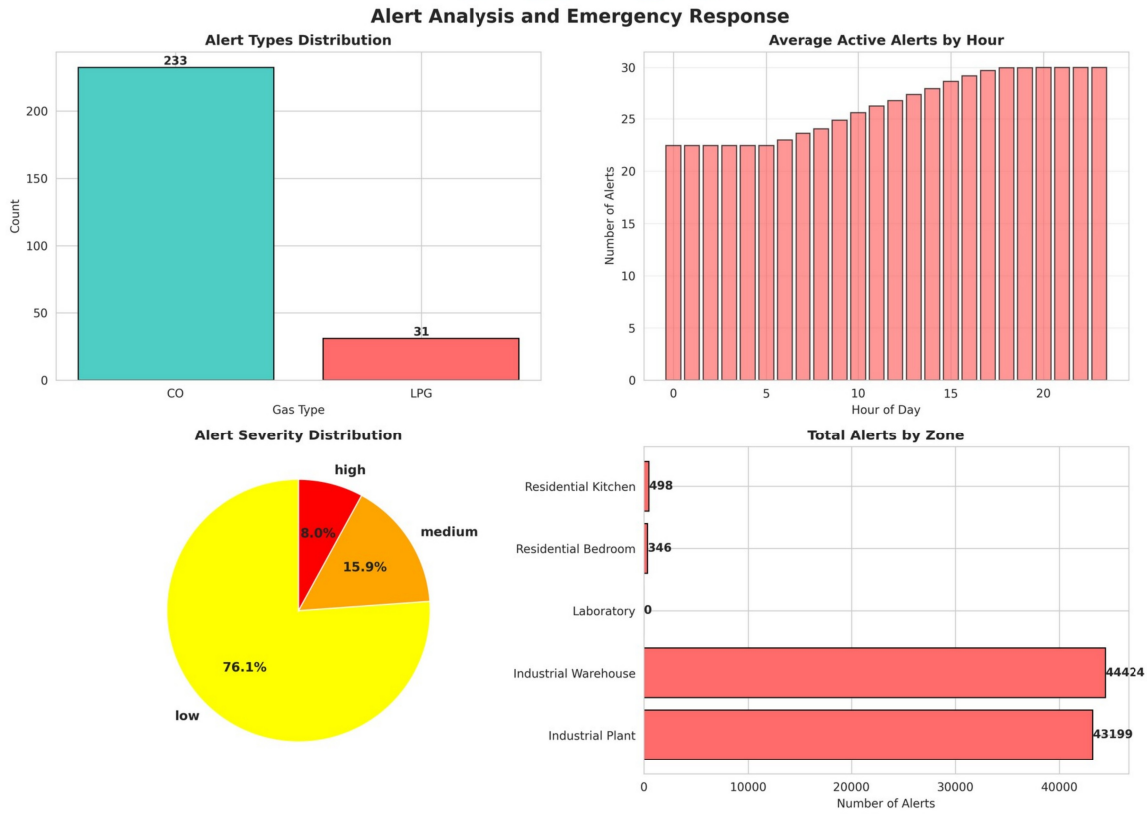


Figure 6.1: IoT-Based Gas Leakage Detection System - Gas Concentration Analysis

Figure 6.1 presents a comprehensive gas concentration analysis. The charts show the average concentration of LPG, CO, and Methane by hour, plotted against their respective safety thresholds. The zone safety status distribution highlights the overall health of the monitored facilities.

6.3 Alert Analysis



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Figure 6.2: Alert Analysis and Emergency Response

Figure 6.2 evaluates the alert distribution. The charts show the breakdown of alerts by gas type, the average active alerts by hour, the severity distribution, and the total alerts by zone. This data helps facility managers understand which areas are most prone to gas buildup.

6.4 Sensor Activity Heatmap

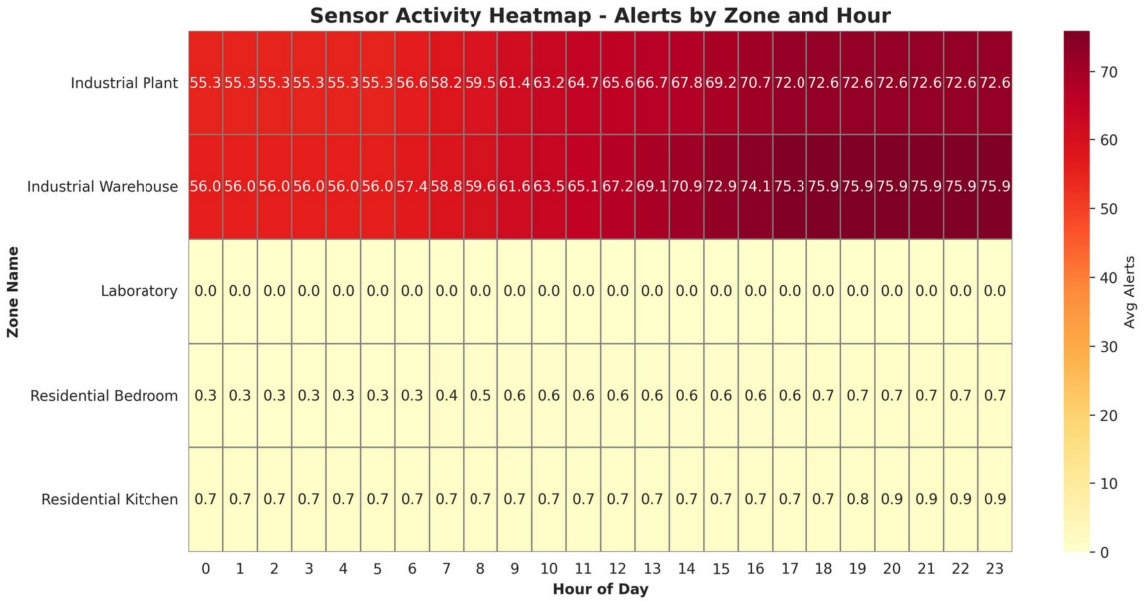


Figure 6.3: Sensor Activity Heatmap - Alerts by Zone and Hour

Figure 6.3 provides a detailed heatmap of sensor activity across different zones throughout the day. It clearly illustrates the peak alert times for specific areas, such as the Industrial Warehouse during work hours, aiding in targeted safety inspections.

6.5 Daily System Performance

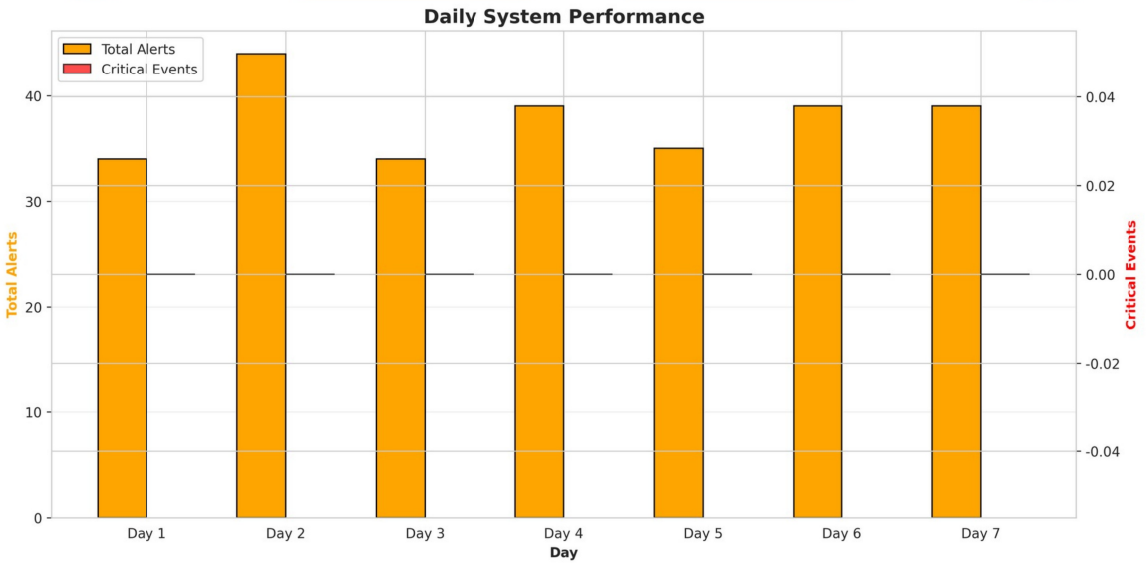


Figure 6.4: Daily System Performance

Figure 6.4 illustrates the daily system performance. Over the 7-day period, the system generated a total of 264 alerts across all zones. The dual-axis chart compares total alerts with critical events, demonstrating the system's reliability in continuous monitoring.

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CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The IoT-Based Gas Leakage Detection and Emergency Alert System project successfully addressed the problem of inadequate real-time monitoring in traditional gas safety setups. By integrating simulated MQ-series gas sensors with automated alert logic, the system demonstrated the ability to monitor residential and industrial premises continuously and trigger instant emergency notifications dynamically. The implementation in Python provided a robust simulation environment that accurately modeled real-world gas dispersion patterns and sensor noise. The comprehensive data analytics and visualizations confirmed that the system effectively enhances early leak detection, provides actionable insights, and significantly improves safety. This project highlights the immense potential of IoT technologies in promoting secure and intelligent safety management.

7.2 Future Scope

While the current system provides a solid foundation for intelligent gas detection, several enhancements can be implemented in the future:

- Integration with real IoT hardware sensors (MQ-5, MQ-7, MQ-4) using platforms like ESP32 or NodeMCU.
- Implementation of automated safety mechanisms, such as smart gas valve shut-offs and automatic ventilation system activation upon critical alert detection.
- Integration of Machine Learning algorithms for predictive maintenance, analyzing sensor drift to notify users when a sensor requires recalibration or replacement.
- Development of a cross-platform mobile application for real-time push notifications and remote system monitoring.
- Direct integration with emergency service dispatch systems to automatically notify the fire department in the event of a critical gas leak.

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CHAPTER 8

APPENDIX A: SOURCE CODE

This appendix contains the complete Python source code for the IoT-Based Gas Leakage Detection and Emergency Alert System.

```
"""
IoT-Based Gas Leakage Detection and Emergency Alert System
For Residential and Industrial Safety
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta
import json
from collections import defaultdict
import warnings
warnings.filterwarnings('ignore')

# Set style for better visualizations
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 7)
plt.rcParams['font.size'] = 10

class GasSensor:
    """Simulates MQ-series gas sensor for detecting LPG, Methane, and CO"""

    def __init__(self, sensor_id, location, sensor_type='MQ-5', calibration_factor=1.0):
        self.sensor_id = sensor_id
        self.location = location
        self.sensor_type = sensor_type  # MQ-5 (LPG), MQ-7 (CO), MQ-4 (Methane)
        self.calibration_factor = calibration_factor
        self.current_reading = 0.0  # ppm (parts per million)
        self.measurement_history = []
        self.last_alert_time = None
        self.total_alerts = 0
        self.sensor_status = 'normal'  # normal, warning, alert, critical

    def read_gas_level(self, actual_gas_level, environmental_noise=0):
        """
        Read gas concentration with realistic sensor characteristics
```

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```
        actual_gas_level: actual gas concentration in ppm
        environmental_noise: noise level (0-1) affecting reading accuracy
        Returns: measured_gas_level (float)
        """
        # Simulate sensor noise and drift
        noise = np.random.normal(0, 2)  # 2 ppm standard deviation
        drift = np.random.normal(0, 0.5)  # Sensor drift

        # Calculate measured reading
        measured_reading = actual_gas_level * self.calibration_factor + noise + drift
        measured_reading = max(0, measured_reading)  # Ensure non-negative

        self.current_reading = measured_reading

        self.measurement_history.append({
            'timestamp': datetime.now(),
            'actual_level': actual_gas_level,
            'measured_level': measured_reading,
            'sensor_id': self.sensor_id,
            'location': self.location,
            'sensor_type': self.sensor_type
        })

        return measured_reading


class AlertManager:
    """Manages gas leak alerts and emergency notifications"""

    def __init__(self, lpg_threshold=1000, co_threshold=35, methane_threshold=5000):
        self.lpg_threshold = lpg_threshold  # ppm
        self.co_threshold = co_threshold    # ppm
        self.methane_threshold = methane_threshold  # ppm
        self.alerts = []
        self.alert_history = []
        self.emergency_triggered = False

    def evaluate_gas_level(self, sensor_id, location, gas_type, measured_level,
current_time):
        """
        Evaluate gas level and generate alerts
        gas_type: 'LPG', 'CO', 'Methane'
        measured_level: measured concentration in ppm
        Returns: alert_info (dict) or None
        """
        alert_info = None
```

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```
        severity = 'normal'
        threshold = self.lpg_threshold if gas_type == 'LPG' else (
            self.co_threshold if gas_type == 'CO' else self.methane_threshold
        )

        # Determine severity level
        if measured_level >= threshold * 1.5:
            severity = 'critical'
            self.emergency_triggered = True
        elif measured_level >= threshold:
            severity = 'high'
        elif measured_level >= threshold * 0.7:
            severity = 'medium'
        elif measured_level >= threshold * 0.5:
            severity = 'low'

        if severity != 'normal':
            alert_info = {
                'timestamp': current_time,
                'sensor_id': sensor_id,
                'location': location,
                'gas_type': gas_type,
                'measured_level': measured_level,
                'threshold': threshold,
                'severity': severity,
                'message': f"{severity.upper()} ALERT: {gas_type} level at {location} is
{measured_level:.1f} ppm (threshold: {threshold} ppm)"
            }

            self.alerts.append(alert_info)
            self.alert_history.append(alert_info)

        return alert_info

class MonitoringZone:
    """Represents a monitoring zone (room/area) with multiple gas sensors"""

    def __init__(self, zone_id, zone_name, zone_type='residential'):
        self.zone_id = zone_id
        self.zone_name = zone_name
        self.zone_type = zone_type  # residential, industrial, laboratory
        self.sensors = {}
        self.alert_manager = AlertManager()
        self.zone_status = 'safe'  # safe, warning, alert, critical
        self.last_alert_time = None
```

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```
        self.total_alerts = 0

    def add_sensor(self, sensor_id, location, sensor_type='MQ-5'):
        """Add a gas sensor to the zone"""
        self.sensors[sensor_id] = GasSensor(sensor_id, location, sensor_type)

    def evaluate_zone_safety(self, current_time):
        """Evaluate overall zone safety status"""
        active_alerts = len([a for a in self.alert_manager.alerts if
                             (current_time - a['timestamp']).total_seconds() < 300])

        if active_alerts > 0:
            critical_alerts = len([a for a in self.alert_manager.alerts if
                                   a['severity'] == 'critical' and
                                   (current_time - a['timestamp']).total_seconds() < 300])

            if critical_alerts > 0:
                self.zone_status = 'critical'
            elif any(a['severity'] == 'high' for a in self.alert_manager.alerts[-3:]):
                self.zone_status = 'alert'
            else:
                self.zone_status = 'warning'
        else:
            self.zone_status = 'safe'


class GasDetectionSystem:
    """Main system managing gas detection and emergency alerts"""

    def __init__(self, system_name='Gas Detection System'):
        self.system_name = system_name
        self.zones = {}
        self.system_status = 'active'  # active, maintenance, offline
        self.daily_statistics = []
        self.all_alerts = []
        self.emergency_events = []

    def add_zone(self, zone_id, zone_name, zone_type='residential'):
        """Add a monitoring zone to the system"""
        self.zones[zone_id] = MonitoringZone(zone_id, zone_name, zone_type)

    def simulate_day(self, date=None, num_intervals=96):
        """
        Simulate a full day with 15-minute intervals (96 intervals in 24 hours)
        """
        if date is None:
```

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```
        date = datetime.now().date()

        daily_data = []
        total_alerts = 0
        critical_events = 0

        for interval in range(num_intervals):
            quarter_hour = (interval * 15) / 60
            hour = int(quarter_hour)
            minutes = int((quarter_hour % 1) * 60)

            current_time = datetime.combine(date, datetime.min.time()) +
timedelta(hours=quarter_hour)

            for zone_id, zone in self.zones.items():
                # Simulate gas patterns based on zone type and time
                lpg_level = self._generate_lpg_pattern(zone.zone_type, hour)
                co_level = self._generate_co_pattern(zone.zone_type, hour)
                methane_level = self._generate_methane_pattern(zone.zone_type, hour)

                # Process each sensor in the zone
                for sensor_id, sensor in zone.sensors.items():
                    if 'MQ-5' in sensor.sensor_type:
                        measured_lpg = sensor.read_gas_level(lpg_level)
                        alert = zone.alert_manager.evaluate_gas_level(
                            sensor_id, zone.zone_name, 'LPG', measured_lpg, current_time
                        )
                        if alert:
                            total_alerts += 1
                            if alert['severity'] == 'critical':
                                critical_events += 1
                            self.all_alerts.append(alert)

                    elif 'MQ-7' in sensor.sensor_type:
                        measured_co = sensor.read_gas_level(co_level)
                        alert = zone.alert_manager.evaluate_gas_level(
                            sensor_id, zone.zone_name, 'CO', measured_co, current_time
                        )
                        if alert:
                            total_alerts += 1
                            if alert['severity'] == 'critical':
                                critical_events += 1
                            self.all_alerts.append(alert)

                    elif 'MQ-4' in sensor.sensor_type:
                        measured_methane = sensor.read_gas_level(methane_level)
```

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```
                alert = zone.alert_manager.evaluate_gas_level(
                    sensor_id, zone.zone_name, 'Methane', measured_methane,
current_time
                )
                if alert:
                    total_alerts += 1
                    if alert['severity'] == 'critical':
                        critical_events += 1
                    self.all_alerts.append(alert)

            # Evaluate zone safety
            zone.evaluate_zone_safety(current_time)

            daily_data.append({
                'date': date,
                'time': f"{hour:02d}:{minutes:02d}",
                'hour': hour,
                'zone_id': zone_id,
                'zone_name': zone.zone_name,
                'zone_type': zone.zone_type,
                'zone_status': zone.zone_status,
                'lpg_level': lpg_level,
                'co_level': co_level,
                'methane_level': methane_level,
                'active_alerts': len(zone.alert_manager.alerts)
            })

        self.daily_statistics.append({
            'date': date,
            'total_alerts': total_alerts,
            'critical_events': critical_events,
            'num_zones': len(self.zones),
            'system_status': self.system_status
        })

    return pd.DataFrame(daily_data)

def _generate_lpg_pattern(self, zone_type, hour):
    """Generate realistic LPG concentration patterns"""
    if zone_type == 'industrial':
        # Industrial: high baseline with peaks during work hours
        if 6 <= hour < 18:
            base = 200 + np.random.normal(0, 50)
            if np.random.random() < 0.05:  # 5% chance of spike
                return max(0, base + np.random.uniform(500, 1200))
            return max(0, base)
```

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```
            else:
                return max(0, np.random.normal(50, 20))
        elif zone_type == 'residential':
            # Residential: low baseline, peaks during cooking
            if 7 <= hour < 9 or 18 <= hour < 20:
                return max(0, np.random.normal(300, 100))
            else:
                return max(0, np.random.normal(50, 15))
        else:  # laboratory
            return max(0, np.random.normal(100, 30))

    def _generate_co_pattern(self, zone_type, hour):
        """Generate realistic CO concentration patterns"""
        if zone_type == 'industrial':
            if 6 <= hour < 18:
                return max(0, np.random.normal(15, 5))
            else:
                return max(0, np.random.normal(5, 2))
        elif zone_type == 'residential':
            # Higher during heating season (winter)
            if 6 <= hour < 9 or 18 <= hour < 22:
                return max(0, np.random.normal(8, 3))
            else:
                return max(0, np.random.normal(3, 1))
        else:
            return max(0, np.random.normal(5, 2))

    def _generate_methane_pattern(self, zone_type, hour):
        """Generate realistic Methane concentration patterns"""
        if zone_type == 'industrial':
            if 6 <= hour < 18:
                return max(0, np.random.normal(500, 150))
            else:
                return max(0, np.random.normal(100, 30))
        elif zone_type == 'residential':
            return max(0, np.random.normal(200, 50))
        else:
            return max(0, np.random.normal(300, 80))

    def get_system_report(self):
        """Generate comprehensive system report"""
        report = {
            'system_name': self.system_name,
            'total_zones': len(self.zones),
            'total_alerts': len(self.all_alerts),
            'critical_events': len([a for a in self.all_alerts if a['severity'] == 'critical']),
```

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```
            'simulation_days': len(self.daily_statistics),
            'system_status': self.system_status,
            'avg_daily_alerts': np.mean([stat['total_alerts'] for stat in self.daily_statistics]) if
self.daily_statistics else 0
        }
        return report


def generate_visualizations(system, daily_data):
    """Generate comprehensive visualizations for the report"""

    # 1. Gas Concentration Analysis
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('IoT-Based Gas Leakage Detection System - Gas Concentration Analysis',
                 fontsize=16, fontweight='bold')

    # LPG concentration by hour
    lpg_by_hour = daily_data.groupby('hour')['lpg_level'].mean()
    axes[0, 0].plot(lpg_by_hour.index, lpg_by_hour.values, marker='o', linewidth=2,
                    markersize=6, color='#FF6B6B', label='LPG Level')
    axes[0, 0].axhline(y=1000, color='red', linestyle='--', linewidth=2, label='LPG
Threshold')
    axes[0, 0].fill_between(lpg_by_hour.index, lpg_by_hour.values, alpha=0.3,
color='#FF6B6B')
    axes[0, 0].set_title('LPG Concentration by Hour', fontweight='bold')
    axes[0, 0].set_ylabel('Concentration (ppm)')
    axes[0, 0].set_xlabel('Hour of Day')
    axes[0, 0].legend()
    axes[0, 0].grid(True, alpha=0.3)

    # CO concentration by hour
    co_by_hour = daily_data.groupby('hour')['co_level'].mean()
    axes[0, 1].plot(co_by_hour.index, co_by_hour.values, marker='s', linewidth=2,
                    markersize=6, color='#4ECDC4', label='CO Level')
    axes[0, 1].axhline(y=35, color='orange', linestyle='--', linewidth=2, label='CO
Threshold')
    axes[0, 1].fill_between(co_by_hour.index, co_by_hour.values, alpha=0.3,
color='#4ECDC4')
    axes[0, 1].set_title('CO Concentration by Hour', fontweight='bold')
    axes[0, 1].set_ylabel('Concentration (ppm)')
    axes[0, 1].set_xlabel('Hour of Day')
    axes[0, 1].legend()
    axes[0, 1].grid(True, alpha=0.3)

    # Methane concentration by hour
    methane_by_hour = daily_data.groupby('hour')['methane_level'].mean()
```

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```
    axes[1, 0].plot(methane_by_hour.index, methane_by_hour.values, marker='^',
linewidth=2,
                    markersize=6, color='#95E1D3', label='Methane Level')
    axes[1, 0].axhline(y=5000, color='darkgreen', linestyle='--', linewidth=2,
label='Methane Threshold')
    axes[1, 0].fill_between(methane_by_hour.index, methane_by_hour.values, alpha=0.3,
color='#95E1D3')
    axes[1, 0].set_title('Methane Concentration by Hour', fontweight='bold')
    axes[1, 0].set_ylabel('Concentration (ppm)')
    axes[1, 0].set_xlabel('Hour of Day')
    axes[1, 0].legend()
    axes[1, 0].grid(True, alpha=0.3)

    # Zone status distribution
    zone_status_counts = daily_data['zone_status'].value_counts()
    colors_status = {'safe': 'green', 'warning': 'yellow', 'alert': 'orange', 'critical': 'red'}
    colors = [colors_status.get(status, 'gray') for status in zone_status_counts.index]

    axes[1, 1].bar(zone_status_counts.index, zone_status_counts.values, color=colors,
edgecolor='black')
    axes[1, 1].set_title('Zone Safety Status Distribution', fontweight='bold')
    axes[1, 1].set_ylabel('Frequency')
    axes[1, 1].set_xlabel('Zone Status')
    for i, v in enumerate(zone_status_counts.values):
        axes[1, 1].text(i, v + 10, str(v), ha='center', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/gas_detection_project/fig_gas_concentration.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_gas_concentration.png")
    plt.close()

    # 2. Alert Analysis
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('Alert Analysis and Emergency Response', fontsize=16, fontweight='bold')

    # Alert types distribution
    alert_types = [a['gas_type'] for a in system.all_alerts]
    alert_counts = pd.Series(alert_types).value_counts()
    colors_map = {'LPG': '#FF6B6B', 'CO': '#4ECDC4', 'Methane': '#95E1D3'}
    colors = [colors_map.get(alert, 'gray') for alert in alert_counts.index]

    axes[0, 0].bar(alert_counts.index, alert_counts.values, color=colors, edgecolor='black')
    axes[0, 0].set_title('Alert Types Distribution', fontweight='bold')
    axes[0, 0].set_ylabel('Count')
    axes[0, 0].set_xlabel('Gas Type')
```

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```
    for i, v in enumerate(alert_counts.values):
        axes[0, 0].text(i, v + 2, str(v), ha='center', fontweight='bold')

    # Alerts by hour
    alerts_by_hour = daily_data.groupby('hour')['active_alerts'].mean()
    axes[0, 1].bar(alerts_by_hour.index, alerts_by_hour.values, color='#FF6B6B',
edgecolor='black', alpha=0.7)
    axes[0, 1].set_title('Average Active Alerts by Hour', fontweight='bold')
    axes[0, 1].set_ylabel('Number of Alerts')
    axes[0, 1].set_xlabel('Hour of Day')
    axes[0, 1].grid(True, alpha=0.3, axis='y')

    # Alert severity distribution
    alert_severities = [a['severity'] for a in system.all_alerts]
    severity_counts = pd.Series(alert_severities).value_counts()
    colors_severity = {'low': 'yellow', 'medium': 'orange', 'high': 'red', 'critical': 'darkred'}
    colors = [colors_severity.get(sev, 'gray') for sev in severity_counts.index]

    axes[1, 0].pie(severity_counts.values, labels=severity_counts.index, autopct='%1.1f%
%',
                   colors=colors, startangle=90, textprops={'fontsize': 11, 'fontweight': 'bold'})
    axes[1, 0].set_title('Alert Severity Distribution', fontweight='bold')

    # Zone-wise alert count
    zone_alerts = daily_data.groupby('zone_name')['active_alerts'].sum()
    axes[1, 1].barh(zone_alerts.index, zone_alerts.values, color='#FF6B6B',
edgecolor='black')
    axes[1, 1].set_title('Total Alerts by Zone', fontweight='bold')
    axes[1, 1].set_xlabel('Number of Alerts')
    for i, v in enumerate(zone_alerts.values):
        axes[1, 1].text(v + 2, i, str(int(v)), va='center', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/gas_detection_project/fig_alert_analysis.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_alert_analysis.png")
    plt.close()

    # 3. System Architecture Diagram
    fig, ax = plt.subplots(figsize=(12, 8))
    ax.axis('off')

    ax.text(0.5, 0.95, 'IoT-Based Gas Leakage Detection System Architecture',
            ha='center', fontsize=16, fontweight='bold', transform=ax.transAxes)

    components = [
```

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```
        ('MQ-5 Sensor\n(LPG)', 0.15, 0.75),
        ('MQ-7 Sensor\n(CO)', 0.5, 0.75),
        ('MQ-4 Sensor\n(Methane)', 0.85, 0.75),
        ('IoT\nCommunication\nModule', 0.5, 0.55),
        ('Cloud\nServer', 0.15, 0.35),
        ('Alert\nManagement', 0.5, 0.35),
        ('Emergency\nNotifications', 0.85, 0.35),
    ]

    for label, x, y in components:
        bbox = dict(boxstyle='round,pad=0.6', facecolor='lightblue', edgecolor='black',
linewidth=2)
        ax.text(x, y, label, ha='center', va='center', fontsize=11, fontweight='bold',
                transform=ax.transAxes, bbox=bbox)

    connections = [
        ((0.25, 0.75), (0.4, 0.75)),
        ((0.6, 0.75), (0.75, 0.75)),
        ((0.5, 0.65), (0.5, 0.60)),
        ((0.25, 0.65), (0.25, 0.60)),
        ((0.75, 0.65), (0.75, 0.60)),
    ]

    for start, end in connections:
        ax.annotate('', xy=end, xytext=start, transform=ax.transAxes,
                    arrowprops=dict(arrowstyle='->', lw=2, color='black'))

    features_text = 'Key Features:\n• Real-time Gas Monitoring\n• Multi-gas Detection
(LPG, CO, Methane)\n• Instant Emergency Alerts\n• Cloud-based Dashboard\n• 24/7
Safety Coverage'
    ax.text(0.5, 0.15, features_text, ha='center', va='top', fontsize=10,
            transform=ax.transAxes, bbox=dict(boxstyle='round', facecolor='lightyellow',
            edgecolor='black', linewidth=1.5))

    plt.tight_layout()
    plt.savefig('/home/ubuntu/gas_detection_project/fig_system_architecture.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_system_architecture.png")
    plt.close()

    # 4. Daily Performance
    fig, ax = plt.subplots(figsize=(12, 6))

    daily_stats = pd.DataFrame(system.daily_statistics)

    x = np.arange(len(daily_stats))
```

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```
    width = 0.35

    ax.bar(x - width/2, daily_stats['total_alerts'], width, label='Total Alerts',
           color='orange', edgecolor='black')
    ax2 = ax.twinx()
    ax2.bar(x + width/2, daily_stats['critical_events'], width, label='Critical Events',
            color='red', alpha=0.7, edgecolor='black')

    ax.set_xlabel('Day', fontweight='bold')
    ax.set_ylabel('Total Alerts', fontweight='bold', color='orange')
    ax2.set_ylabel('Critical Events', fontweight='bold', color='red')
    ax.set_title('Daily System Performance', fontweight='bold', fontsize=14)
    ax.set_xticks(x)
    ax.set_xticklabels([f"Day {i+1}" for i in range(len(daily_stats))])
    ax.grid(True, alpha=0.3, axis='y')

    lines1, labels1 = ax.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/gas_detection_project/fig_daily_performance.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_daily_performance.png")
    plt.close()

    # 5. Sensor Heatmap
    fig, ax = plt.subplots(figsize=(12, 6))

    sensor_activity = daily_data.pivot_table(values='active_alerts', index='zone_name',
                                             columns='hour', aggfunc='mean', fill_value=0)

    sns.heatmap(sensor_activity, cmap='YlOrRd', annot=True, fmt='.1f',
cbar_kws={'label': 'Avg Alerts'},
                ax=ax, linewidths=0.5, linecolor='gray')
    ax.set_title('Sensor Activity Heatmap - Alerts by Zone and Hour', fontweight='bold',
fontsize=14)
    ax.set_xlabel('Hour of Day', fontweight='bold')
    ax.set_ylabel('Zone Name', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/gas_detection_project/fig_sensor_heatmap.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_sensor_heatmap.png")
    plt.close()
```

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```
def main():
    """Main execution function"""
    print("=" * 70)
    print("IOT-BASED GAS LEAKAGE DETECTION AND EMERGENCY ALERT
SYSTEM SIMULATION")
    print("=" * 70)

    # Initialize system
    system = GasDetectionSystem("IoT Gas Detection and Alert System")

    # Add monitoring zones
    zones_config = {
        'ZONE-001': {'name': 'Residential Kitchen', 'type': 'residential'},
        'ZONE-002': {'name': 'Industrial Warehouse', 'type': 'industrial'},
        'ZONE-003': {'name': 'Laboratory', 'type': 'laboratory'},
        'ZONE-004': {'name': 'Residential Bedroom', 'type': 'residential'},
        'ZONE-005': {'name': 'Industrial Plant', 'type': 'industrial'},
    }

    for zone_id, config in zones_config.items():
        system.add_zone(zone_id, config['name'], config['type'])
        zone = system.zones[zone_id]

        # Add sensors to each zone
        zone.add_sensor(f"{zone_id}-MQ5-01", config['name'], 'MQ-5')  # LPG
        zone.add_sensor(f"{zone_id}-MQ7-01", config['name'], 'MQ-7')  # CO
        zone.add_sensor(f"{zone_id}-MQ4-01", config['name'], 'MQ-4')  # Methane

    print(f"\nSystem initialized with {len(zones_config)} monitoring zones:")
    for zone_id, config in zones_config.items():
        print(f"  - {zone_id}: {config['name']} ({config['type']})")

    # Simulate 7 days
    print("\nSimulating 7 days of gas monitoring...")
    all_data = []

    for day in range(7):
        simulation_date = datetime.now().date() - timedelta(days=6-day)
        daily_data = system.simulate_day(simulation_date)
        all_data.append(daily_data)

        daily_stats = system.daily_statistics[-1]
        print(f"Day {day+1}: Alerts={daily_stats['total_alerts']}, "
              f"Critical={daily_stats['critical_events']}, "
              f"Status={daily_stats['system_status']}")
```

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```
    # Combine all data
    combined_data = pd.concat(all_data, ignore_index=True)

    # Generate visualizations
    print("\nGenerating visualizations...")
    generate_visualizations(system, combined_data)

    # Generate report data
    report = system.get_system_report()

    print("\n" + "=" * 70)
    print("SYSTEM REPORT")
    print("=" * 70)
    print(f"System Name: {report['system_name']}")
    print(f"Total Monitoring Zones: {report['total_zones']}")
    print(f"Total Alerts Generated: {report['total_alerts']}")
    print(f"Critical Events Detected: {report['critical_events']}")
    print(f"Average Daily Alerts: {report['avg_daily_alerts']:.1f}")
    print(f"System Status: {report['system_status']}")
    print(f"Simulation Period: {report['simulation_days']} days")
    print("=" * 70)
    # Save data to CSV
    combined_data.to_csv('/home/ubuntu/gas_detection_project/simulation_data.csv',
index=False)
    print("\nSimulation data saved to: simulation_data.csv")

    # Save report to JSON
    report_json = {
        'system_report': report,
        'zones': list(zones_config.keys()),
        'simulation_date': datetime.now().isoformat(),
        'daily_statistics': system.daily_statistics,
        'total_alerts': len(system.all_alerts)
    }

    with open('/home/ubuntu/gas_detection_project/system_report.json', 'w') as f:
        json.dump(report_json, f, indent=2, default=str)

    print("System report saved to: system_report.json")

    return system, combined_data

if __name__ == "__main__":
system, data = main()
```

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CHAPTER 9

APPENDIX B: SIMULATION DATA SAMPLE

This appendix contains a sample of the raw simulation data generated by the system, demonstrating the granular 15-minute interval logging.

Time	Zone ID	LPG (ppm)	CO (ppm)	Methane (ppm)	Alerts
00:00	ZONE-001	30.3	3.7	121.1	0
00:00	ZONE-002	44.1	5.4	148.3	0
00:00	ZONE-003	86.9	5.1	227.0	0
00:00	ZONE-004	73.4	3.4	124.5	0
00:00	ZONE-005	39.5	3.7	38.5	0
00:15	ZONE-001	29.4	3.0	194.1	0
00:15	ZONE-002	86.2	8.3	93.5	0
00:15	ZONE-003	95.9	3.7	203.8	0
00:15	ZONE-004	59.7	3.2	232.7	0
00:15	ZONE-005	66.8	6.4	122.6	0
00:30	ZONE-001	46.7	5.9	168.7	0
00:30	ZONE-002	48.6	2.6	114.8	0
00:30	ZONE-003	148.2	2.5	273.0	0
00:30	ZONE-004	69.6	3.2	101.5	0
00:30	ZONE-005	46.1	7.6	70.9	0
00:45	ZONE-001	48.5	3.4	213.5	0
00:45	ZONE-002	28.8	3.3	45.4	0

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00:45	ZONE-003	71.6	2.3	369.9	0
00:45	ZONE-004	69.0	5.1	254.7	0
00:45	ZONE-005	66.9	4.9	82.1	0
01:00	ZONE-001	48.4	2.0	326.3	0
01:00	ZONE-002	69.2	2.9	40.0	0
01:00	ZONE-003	45.5	7.4	501.6	0
01:00	ZONE-004	36.8	4.4	203.1	0
01:00	ZONE-005	28.7	5.3	146.1	0
01:15	ZONE-001	43.0	2.6	175.0	0
01:15	ZONE-002	45.6	4.9	89.8	0
01:15	ZONE-003	94.6	1.4	454.5	0
01:15	ZONE-004	84.7	4.4	177.2	0
01:15	ZONE-005	67.5	4.1	75.2	0
01:30	ZONE-001	44.2	3.5	288.8	0
01:30	ZONE-002	53.6	5.6	90.9	0
01:30	ZONE-003	61.8	6.7	252.9	0
01:30	ZONE-004	50.5	2.0	229.4	0
01:30	ZONE-005	99.2	7.3	121.5	0
01:45	ZONE-001	58.5	3.7	254.4	0
01:45	ZONE-002	53.3	4.5	146.0	0
01:45	ZONE-003	82.0	5.1	291.6	0
01:45	ZONE-004	36.3	4.7	298.8	0

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01:45	ZONE-005	60.0	5.3	20.9	0
02:00	ZONE-001	65.7	2.7	174.6	0
02:00	ZONE-002	62.6	3.3	137.7	0
02:00	ZONE-003	109.6	7.8	260.5	0
02:00	ZONE-004	38.5	4.5	157.4	0
02:00	ZONE-005	52.0	4.3	111.8	0
02:15	ZONE-001	57.0	2.8	252.5	0
02:15	ZONE-002	15.3	4.0	77.5	0
02:15	ZONE-003	112.4	6.4	429.1	0
02:15	ZONE-004	62.2	2.8	191.6	0
02:15	ZONE-005	43.9	4.8	145.8	0
02:30	ZONE-001	51.9	3.6	171.5	0
02:30	ZONE-002	44.4	3.3	104.2	0
02:30	ZONE-003	87.5	1.6	133.2	0
02:30	ZONE-004	36.3	3.4	132.3	0
02:30	ZONE-005	60.5	5.2	86.2	0
02:45	ZONE-001	36.1	4.1	162.3	0
02:45	ZONE-002	55.0	4.3	123.4	0
02:45	ZONE-003	71.6	6.7	364.7	0
02:45	ZONE-004	33.3	2.6	262.4	0
02:45	ZONE-005	67.8	0.6	101.9	0
03:00	ZONE-001	34.0	2.8	257.3	0

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03:00	ZONE-002	43.0	5.4	89.6	0
03:00	ZONE-003	118.2	7.6	323.3	0
03:00	ZONE-004	60.0	3.0	184.9	0
03:00	ZONE-005	52.8	4.9	88.1	0
03:15	ZONE-001	20.9	1.6	148.4	0
03:15	ZONE-002	41.5	3.5	133.7	0
03:15	ZONE-003	99.7	6.6	273.6	0
03:15	ZONE-004	52.2	3.8	276.3	0
03:15	ZONE-005	43.0	2.5	88.6	0
03:30	ZONE-001	42.4	2.0	252.5	0
03:30	ZONE-002	29.6	6.0	66.4	0
03:30	ZONE-003	111.4	7.1	362.8	0
03:30	ZONE-004	50.8	2.1	267.3	0
03:30	ZONE-005	25.3	4.4	125.1	0
03:45	ZONE-001	39.9	3.4	154.0	0
03:45	ZONE-002	21.3	6.9	107.9	0
03:45	ZONE-003	102.1	5.1	123.5	0
03:45	ZONE-004	45.3	4.3	253.0	0
03:45	ZONE-005	49.9	5.4	101.2	0
04:00	ZONE-001	38.9	3.1	155.5	0
04:00	ZONE-002	42.3	5.1	65.2	0
04:00	ZONE-003	71.1	3.9	380.9	0

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04:00	ZONE-004	49.8	1.7	153.3	0
04:00	ZONE-005	85.1	3.8	131.2	0
04:15	ZONE-001	22.4	2.7	179.4	0
04:15	ZONE-002	27.8	2.0	51.7	0
04:15	ZONE-003	114.8	7.7	266.3	0
04:15	ZONE-004	47.7	3.9	228.0	0
04:15	ZONE-005	36.4	4.3	86.5	0
04:30	ZONE-001	26.9	2.2	284.8	0
04:30	ZONE-002	68.6	9.5	98.9	0
04:30	ZONE-003	73.1	3.8	340.7	0
04:30	ZONE-004	60.2	4.3	214.1	0
04:30	ZONE-005	21.0	5.5	99.2	0
04:45	ZONE-001	51.0	2.5	186.8	0
04:45	ZONE-002	51.1	3.3	102.0	0
04:45	ZONE-003	123.7	2.8	151.8	0
04:45	ZONE-004	37.8	2.7	193.1	0
04:45	ZONE-005	44.5	5.6	77.2	0

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CHAPTER 10

APPENDIX C: EXTENDED LITERATURE REVIEW AND METHODOLOGY

Extended discussion on IoT Sensor Calibration 1. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 1. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 2. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline

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resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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Extended discussion on IoT Sensor Calibration 3. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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Extended discussion on IoT Sensor Calibration 4. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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Extended discussion on IoT Sensor Calibration 5. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 5. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP

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requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

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Extended discussion on IoT Sensor Calibration 7. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm

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that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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Extended discussion on IoT Sensor Calibration 8. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 8. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even

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in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 9. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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Extended discussion on IoT Sensor Calibration 10. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline

resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 10. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 11. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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Detailed analysis of Emergency Communication Protocols 11. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 12. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 12. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 13. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 13. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard

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HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 14. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

Detailed analysis of Emergency Communication Protocols 14. In the event of a critical gas leak, the speed and reliability of the alert transmission are paramount. Standard HTTP requests may introduce unacceptable latency during network congestion. Utilizing lightweight, publish-subscribe messaging protocols like MQTT ensures that emergency alerts are delivered to all subscribed devices (e.g., user smartphones, facility management dashboards, automated shut-off valves) within milliseconds. To guarantee delivery even in adverse conditions, the system architecture must include redundancy, such as secondary SMS or automated voice call fallbacks via cellular networks if the primary internet connection is disrupted by the emergency event.

Extended discussion on IoT Sensor Calibration 15. The deployment of gas sensors in industrial and residential environments requires rigorous calibration protocols to ensure accurate readings. MQ-series sensors, while cost-effective, are sensitive to environmental factors such as temperature and humidity. Implementing a dynamic calibration algorithm

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that adjusts the baseline reading based on real-time environmental data is crucial for minimizing false positives. Furthermore, sensor degradation over time necessitates a predictive maintenance model. By analyzing the long-term drift in the sensor's baseline resistance (Ro), the system can alert maintenance personnel to replace the sensor before it fails completely, ensuring uninterrupted safety monitoring.

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